LAMMPS official dem05-sliptime

Frictional sliding and spin before rolling starts | 29 October 2026

PASS	Four main runs evaluated using the unchanged official selected-time analytic criteria; source tag unvetted.

Official quantities, errors and acceptance limits

Quantity	Reference / limit	CPU dt	GPU dt	CPU dt/2	GPU dt/2
Sliding vx (m/s)	1.24 relative limit 0.01	1.240002 err 1.583e-06 PASS	1.240002 err 1.583e-06 PASS	1.240001 err 1.173e-06 PASS	1.240001 err 1.173e-06 PASS
Sliding omega_y (rad/s)	160 relative limit 0.01	159.9969 err 1.934e-05 PASS	159.9969 err 1.934e-05 PASS	159.9974 err 1.616e-05 PASS	159.9974 err 1.616e-05 PASS

Check time=0.4 s; original step 40000. Half dt uses twice the check step at the same time. Each cell shows the native value and its own absolute/relative error; a reference limit is not a same-code epsilon.

Original bodies and source-to-native mapping

Body	Radius / density	Initial xyz (m)	Initial v (m/s)	Initial spin (rad/s)
sphere	R=0.0025 rho=2700	(0, 0, 0.002499982)	(1.4, 0, 0)	(0, 0, 0)

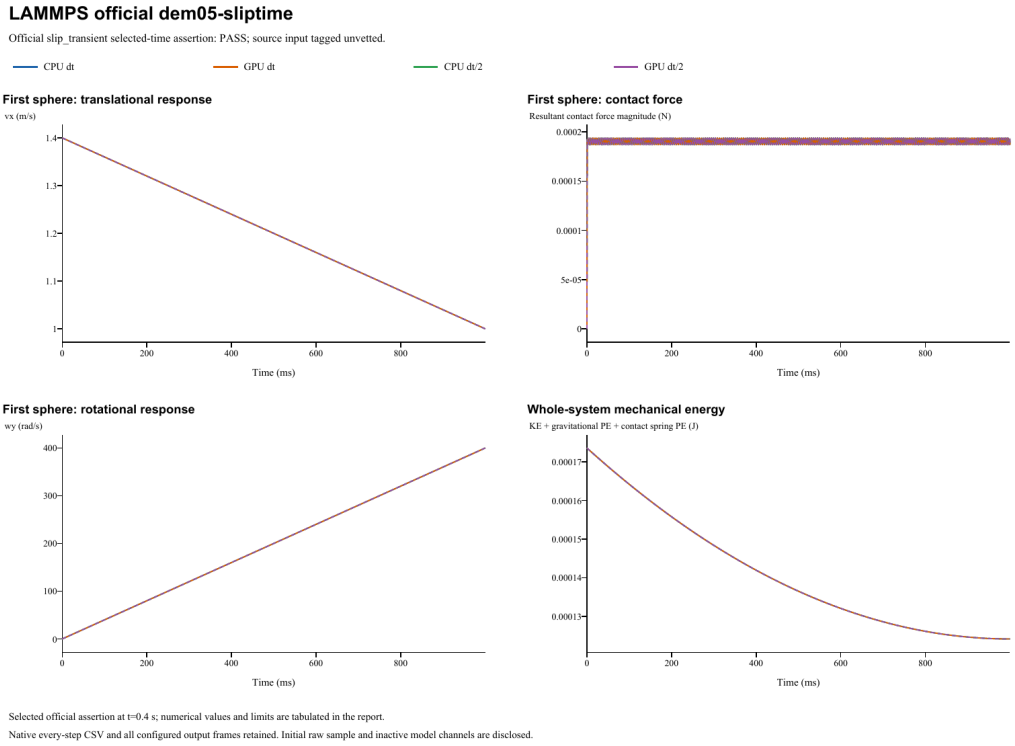
Linear Hooke Fn_elastic=kn*delta, kn=10000 N/m. Automatic damping uses the harmonic material restitution and the active normal/tangential stiffness; stored manual dashpots are inactive. Tangential inputs: kt=2800 N/m (explicit in linear mode), mu=0.4. Rolling/twist coefficients: kr=0, mu_r=0, ktwist=0, mu_twist=0. Source wall spin-slip and torque use R; native uses the shared contact point at R-delta. Maximum source/native contact-speed difference is 7.198e-06 m/s; torque-arm difference is 1.8e-08 m. Official selected-time acceptance does not imply pointwise kinematic identity.

Gravity=(0, 0, -1) m/s^2. Base/half dt=1e-05/5e-06 s; 100000/200000 steps. Source periodic x/y do not affect contact dynamics. Map saved x coordinates modulo original periodic box length0.55m for checkpoint comparison. Source ceiling atz=1m remains inactive but is retained.

Pinned LAMMPS source

Official input | Analytic checker | Parameter mapping.
Commit 29dd3fd7642a6fe9e8b4dd99120df4068b221250; input blob 20f0a60a7ead5d669a420f48008aae1fd26fe928. Official tag: unvetted. LAMMPS was not rerun locally. Physical inputs were not fitted to native results.

LAMMPS native histories and refinement



CPU/GPU and half-step comparison

Quantity	CPU dt error	CPU dt/2 error	Error type / limit
Sliding vx (m/s)	1.583e-06	1.1734e-06	relative: 0.01
Sliding omega_y (rad/s)	1.9338e-05	1.6163e-05	relative: 0.01

CPU/GPU whole-history maximum position / velocity / spin / force differences: 2.72e-14 m / 2.65e-11 m/s / 1.06e-08 rad/s / 3.55e-11 N.

Maximum individual impulse / position-drift / velocity-kick residuals: 1.1e-17 kg m/s / 1.11e-16 m / 6.21e-14 m/s. Gravity is added separately to saved native contact forces.

Raw eventStart samples precede initialization and retain zero force/contact placeholders. Physical geometric preloads are used only for derived bootstrap/impulse audits; native initialization supplies the actual first half-kick. No raw results or initial velocities were changed.

Complete results and verified scope

All 4,004 native frames are retained; main frame counts=[1001, 1001, 1001, 1001]. Every numerical-step particle/contact CSV, four loadable main projects and raw initial samples remain preserved. Output cadence is recorded separately from integration dt. Energy/history diagnostics do not replace the official acceptance quantities.

Run manifest | Analysis | Official metrics | Independent audit | Full native ZIP | Frame hashes

Source checkpoints: Source checkpoint states are supplemental regression diagnostics. Their differences are reported directly; the YAML same-code epsilon is not imposed as an independent cross-software acceptance limit.

A stable helper executable is statically linked to the current native CPU/CUDA libraries. Historical GUI/library byte identity is not asserted. Scope is the active source physics and the selected official criteria; inactive options remain outside this test.